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1.

(**Special note:** Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file [here](#) [🔗](#). If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)

For the following statements, select all that are correct:

☒ The dual program of a linear program is unique.

✔ Correct

☒ The dual program of the dual program of a linear program is the original one.

✔ Correct

☐ If a primal constraint is binding, the corresponding dual variable is positive.

☒ A solution $\bar{x}$ is optimal to a linear program P if and only if there exists another solution $\bar{y}$ that is optimal to the dual program of P and the objective values associated with $\bar{x}$ and $\bar{y}$ are identical.

✔ Correct

☐ None of the above.

1 / 1 point
2.

For the following statements, select all that are correct:

☒ For a linear program, the shadow price of a constraint is unique.

✔ Correct

☐ Suppose that for a linear program the objective value of an optimal solution is z^* . Moreover, the shadow price of a constraint is 3. In this case, if the right-hand side of that constraint is increased by 1, the objective of an optimal solution to the new program will be $z^* + 3$.

☒ For a linear program with a maximization objective function, the shadow price of a greater-than-or-equal-to constraint is nonpositive.

✔ Correct

☐ Suppose that a linear program has been solved, and an optimal basis B^* is obtained. When a new variable is added, B^* still remains optimal.

☐ None of the above.

1 / 1 point
3.

For the following statements, select all that are correct:

☐ A maximum flow problem can be formulated as an assignment problem.

☐ An assignment problem can be formulated as a maximum flow problem.

☐ An assignment problem can be formulated as a shortest path problem.

☐ A shortest path problem can be formulated as a maximum flow problem.

☒ None of the above.

✔ Correct

1 / 1 point
4.

For the following nonlinear programs, select all that are convex programs:

☒ An unconstrained nonlinear program in which a convex function is minimized.

✔ Correct

☐ An unconstrained nonlinear program in which a convex function is maximized.

☐ An unconstrained nonlinear program in which a concave function is minimized.

☒ An unconstrained nonlinear program in which a concave function is maximized.

✔ Correct

☐ None of the above.

1 / 1 point
5.

For the following statements, select all that are correct:

☐ A set is a concave set if it is not a convex set.

☐ The KKT condition is necessary and sufficient for all nonlinear programs.

☒ For the objective value of an optimal solution to a nonlinear program with a maximization objective function, Lagrange relaxation always gives us an upper bound.

✔ Correct

☒ Linear programming duality is a special case of Lagrange duality.

✔ Correct

☐ None of the above.

1 / 1 point